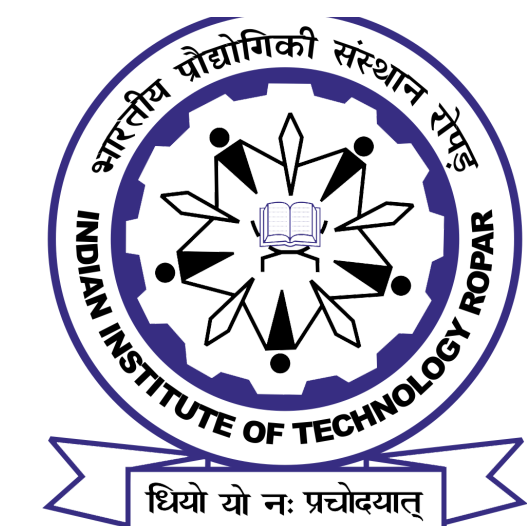




CVIP 2025

CGMap: A Geospatially Aware Deep Learning Framework for Crop Gap Mapping Using UAV

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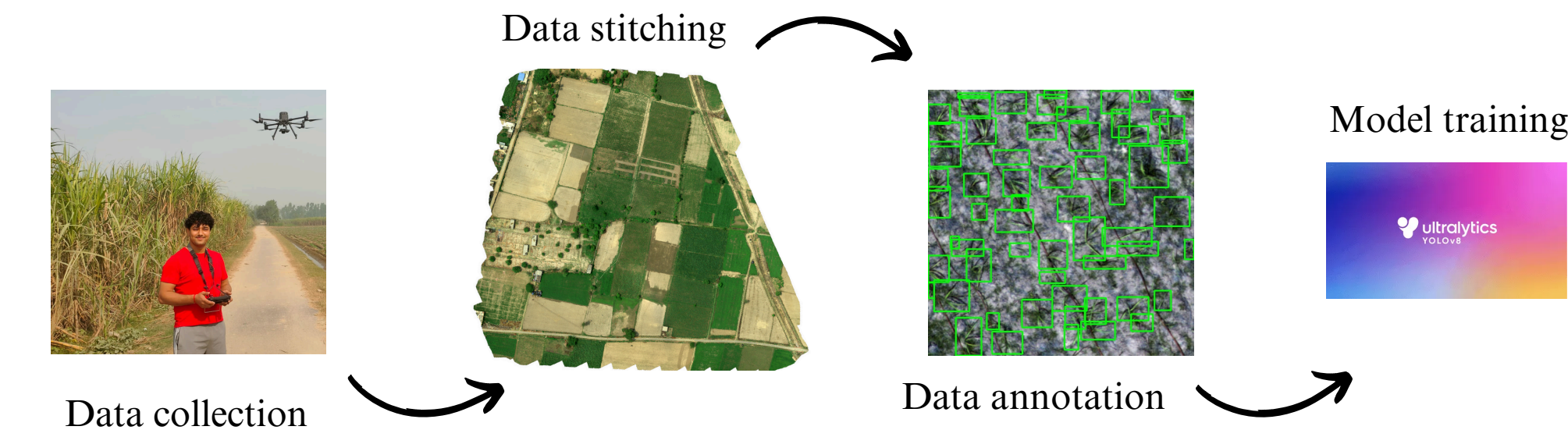


Abstract

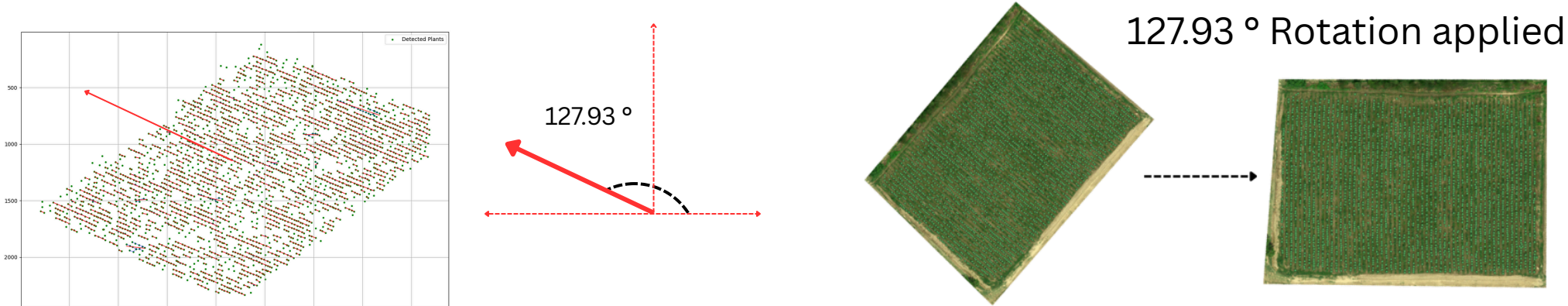
Sugarcane germination in India is still assessed through manual inspections, which are slow, error-prone, and impractical at scale. This work presents CGMap, a geospatially aware, deep-learning-based pipeline that automatically detects early-stage sugarcane saplings from UAV imagery and identifies missing-plant regions (“gaps”). A lightweight object detection model provides plant locations, which are converted into a georeferenced point cloud for spatial analysis. A Minimum Spanning Tree (MST) based orientation normalization technique enables reliable row and column extraction even in irregular Indian farm layouts. Gap detection is then performed using expected intra-row spacing criteria, and all outputs are exported in GIS-ready WKT format for agronomic decision-making. CGMap enables timely transplantation interventions, supports resource-efficient field management, and provides a scalable, interpretable, and generalizable solution for precision agriculture.

Methodology

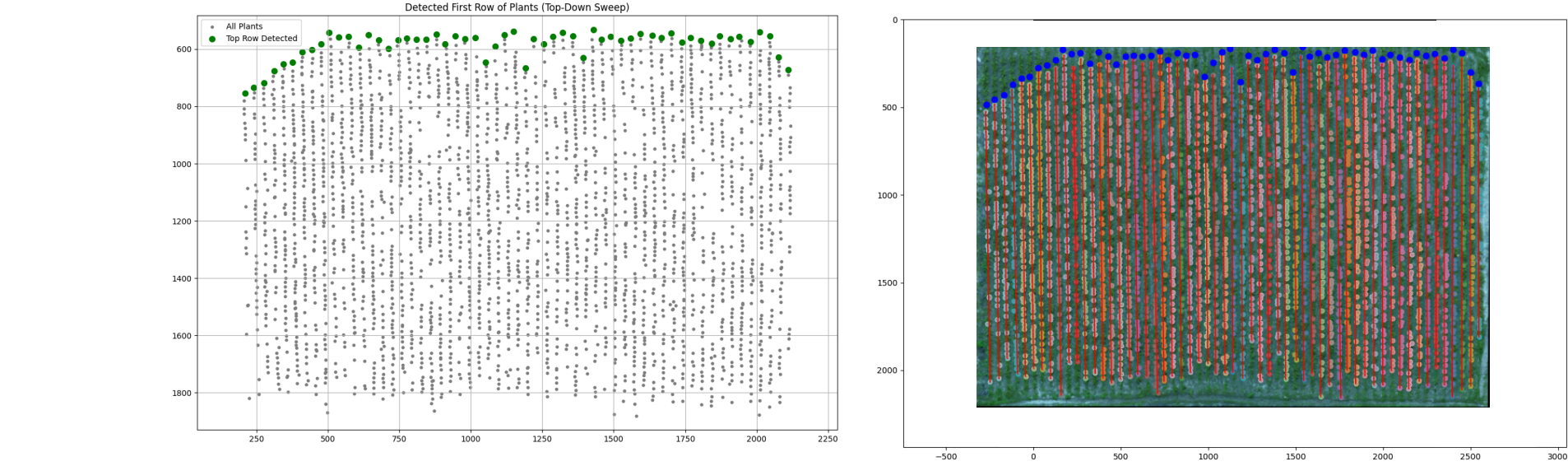
The methodology followed in this study presents an end-to-end geospatial computer-vision pipeline designed to automate early-stage crop emergence assessment using UAV imagery, object detection, and spatial point-cloud reasoning. The workflow begins with the acquisition of high-resolution aerial imagery over selected farms in the Asmoli region of Uttar Pradesh. These farms were surveyed using a UAV platform, and the resulting image sets were processed to generate orthomosaic maps that accurately represent field surface geometry. Farm polygons already provided in coordinate form were used to extract exact field boundaries from the orthomosaic, ensuring clean spatial segmentation before analysis.



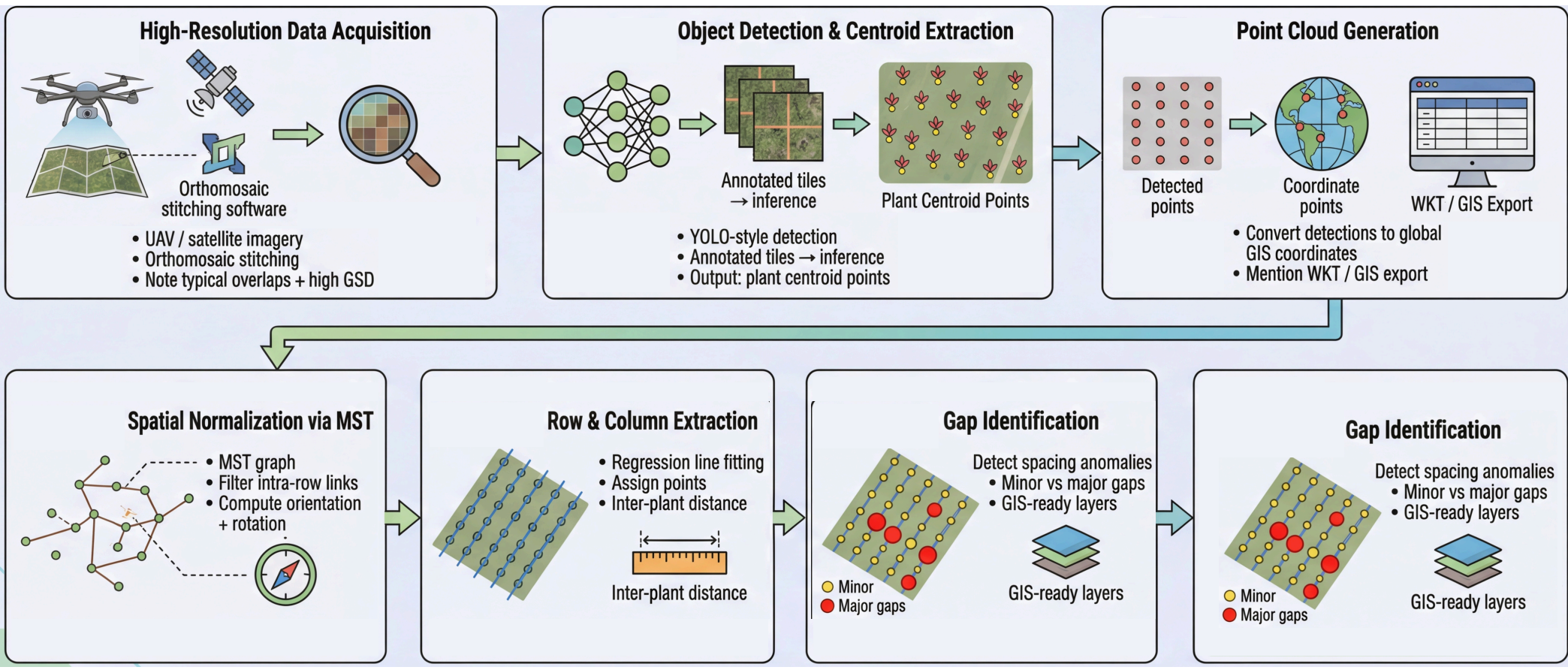
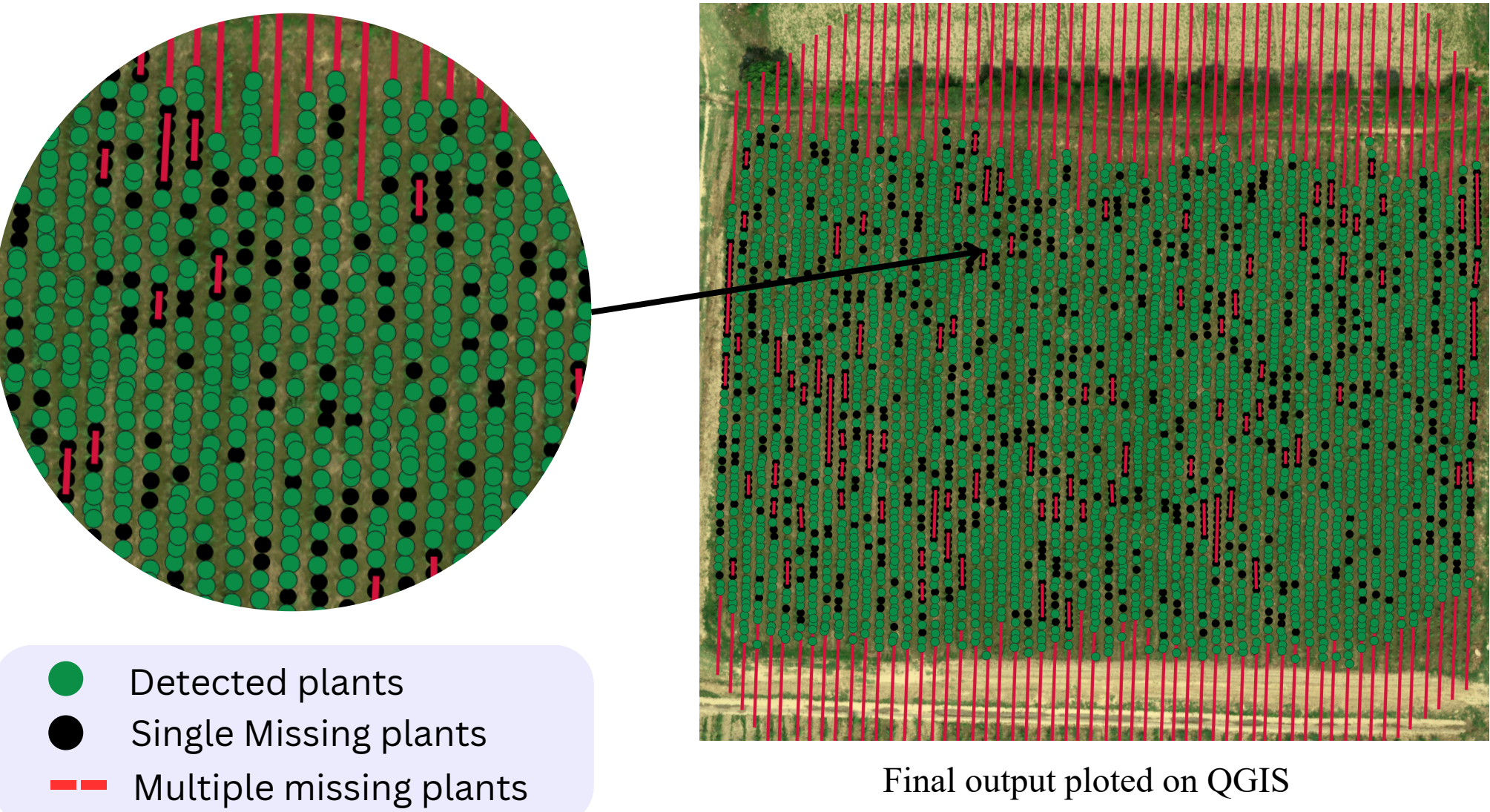
Once the orthomosaic tiles were prepared, a YOLOv8 based detection model was applied to identify individual crop saplings across the field surface. Each detection was converted into a centroid representing a single plant location. These centroids formed a georeferenced point-cloud layer that served as the foundation for all subsequent spatial reasoning steps. To analyze planting structure, a spatial normalization process was performed using a graph-based approach. A fully connected point graph was constructed, and a Minimum Spanning Tree (MST) was derived to infer dominant field orientation. This orientation was used to rotate the point cloud into an aligned frame where plant rows and columns could be reliably extracted. Row and column identification was performed through line-fitting techniques, assigning each plant to its nearest structural line based on perpendicular distance thresholds.



Following structural extraction, inter-plant distances along each column were computed to identify irregular spacing patterns. Gaps were detected by comparing actual distances with expected spacing, distinguishing isolated anomalies from longer consecutive bald patches.

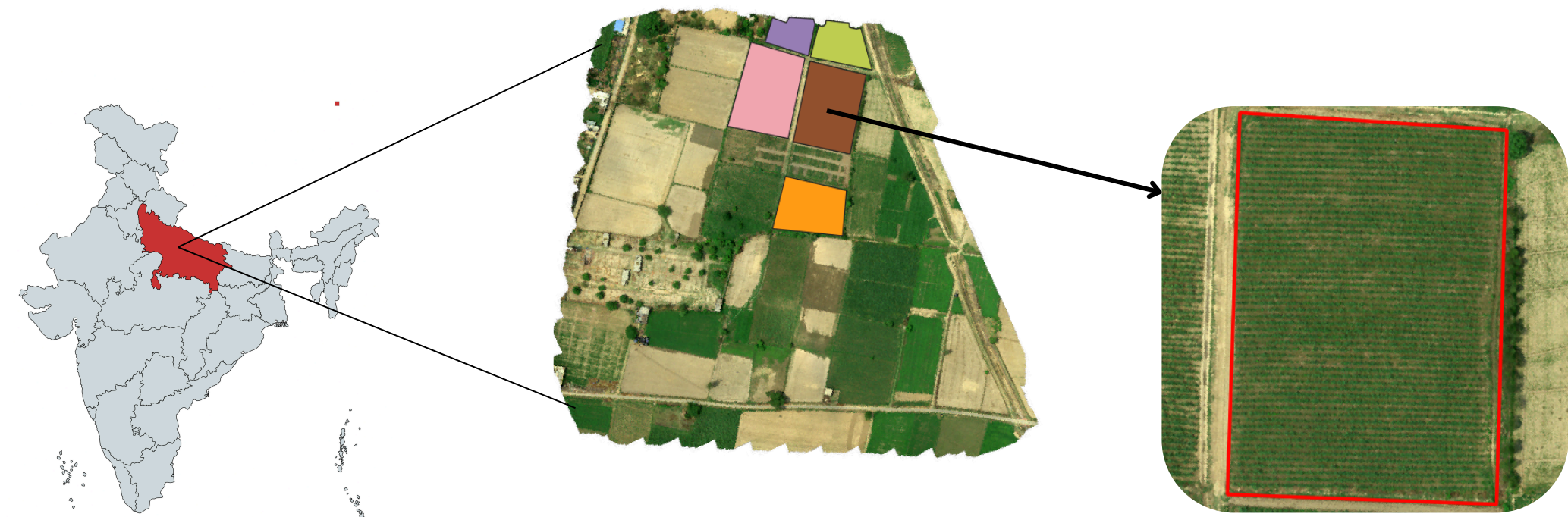


Finally, all outputs including plant points, row lines, and detected gaps were exported in standard GIS compatible formats, enabling seamless integration into agronomy dashboards and decision-support systems.



*AI generated image

Experimental Set-up & Results



Test farm at Asmoli, Uttar Pradesh

The test setup was conducted on sugarcane farms located in the Asmoli region of Uttar Pradesh, where UAV-based orthomosaic imagery was used as the primary data source. Using field boundary coordinates provided beforehand, each farm polygon was extracted from the larger orthomosaic for independent analysis. An object detection model based on the YOLOv8 architecture was then applied to identify individual sugarcane seedlings across the extracted farm regions.

In practical deployments, the runtime of the CGMap pipeline is primarily influenced by whichever object detection model is used, while all subsequent stages operate on a lightweight point-cloud representation that is independent of the model architecture. Once plant locations are obtained, whether from YOLOv8, Faster R-CNN, EfficientDet, or any other detector the pipeline performs only centroid projection, MST-based alignment, row reconstruction, and gap extraction. These operations scale with the number of detected plants rather than with the image resolution, making the spatial analysis component both model-agnostic and computationally efficient.

Machine	FPS	Avg. time (s)
Intel i5-10500	16.10	0.0621
GTX 1660 SUPER	69.82	0.0143
Raspberry Pi 4 (4GB)	0.94	1.0675
Raspberry Pi 5 (8GB)	2.09	0.7212
Intel Xeon W5	19.63	0.0510

On commodity hardware, the complete detection-plus-analysis cycle typically finishes within tens of milliseconds on GPUs and well under a second on mid-range CPUs, with geometric post-processing contributing only a small, stable fraction of the total runtime. This efficiency enables near real-time performance, allowing gap maps to be generated almost immediately after image capture. Even on constrained edge devices such as the Raspberry Pi 4 and Pi 5, the pipeline maintains practical runtimes, underscoring its suitability for scalable field deployment and its flexibility across different detection backbones.

Conclusion

CGMap demonstrates a scalable and interpretable vision-based pipeline for early-stage sugarcane emergence monitoring. By combining UAV imagery, model-independent sapling detection, and point-cloud-driven spatial reasoning, the system reliably maps gaps and supports timely field interventions. Its lightweight, modular design enables deployment in resource-constrained environments and can naturally extend to other crops with similar planting geometries. Future work includes expanding the dataset across seasons and regions, improving robustness, and validating results against ground-truth emergence surveys. Additionally, integrating multispectral cues and temporal growth patterns could further enhance gap detection accuracy. With continued refinement, CGMap can evolve into a comprehensive decision-support tool for precision agriculture across diverse cropping systems. Ultimately, this framework establishes a strong foundation for scalable, data-driven farm management in varied agricultural landscapes.

Acknowledgement

We sincerely thank **CNH Industrial** for their support under **Project Pahal**, a CSR initiative by the company, whose resources and funding were instrumental in enabling this research. We are also grateful to **Keshika Gajbhiye** for her assistance in proofreading the manuscript during its preparation.